

# SMARTX: Porewater Sulfide

May 2025 Samples

2026-06-08

## Code Set up

### run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "May"
sample_year = "2026"
Run_Date = as.Date("05/19/2026", format = "%m/%d/%Y")
Run_by = "Melanie Giessner" #Instrument user
Script_run_by = "Melanie Giessner" #Code user

Run_notes="" #any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7", "Plate8", "Plate9","rrPlate1",")

folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE,pattern = "Plate")
head(Raw_plates)

## [1] "Raw Data/20260519_SMARTX_H2S_Plate1.csv"
## [2] "Raw Data/20260519_SMARTX_H2S_Plate2.csv"
## [3] "Raw Data/20260520_SMARTX_H2S_Plate3.csv"
## [4] "Raw Data/20260520_SMARTX_H2S_Plate4.csv"
## [5] "Raw Data/20260520_SMARTX_H2S_Plate5.csv"
## [6] "Raw Data/20260520_SMARTX_H2S_Plate6.csv"

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
  Plate = plates,
  Run_Date = c("05/19/26","05/19/26","05/20/26","05/20/26","05/20/26","05/20/26","05/21/26","05/21/26",
)

Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, tryFormats = c("%Y-%m-%d", "%m/%d/%y", "%m/%d/%Y"))

# Define the file path for QAQC log file - NO Need to change just check year
```

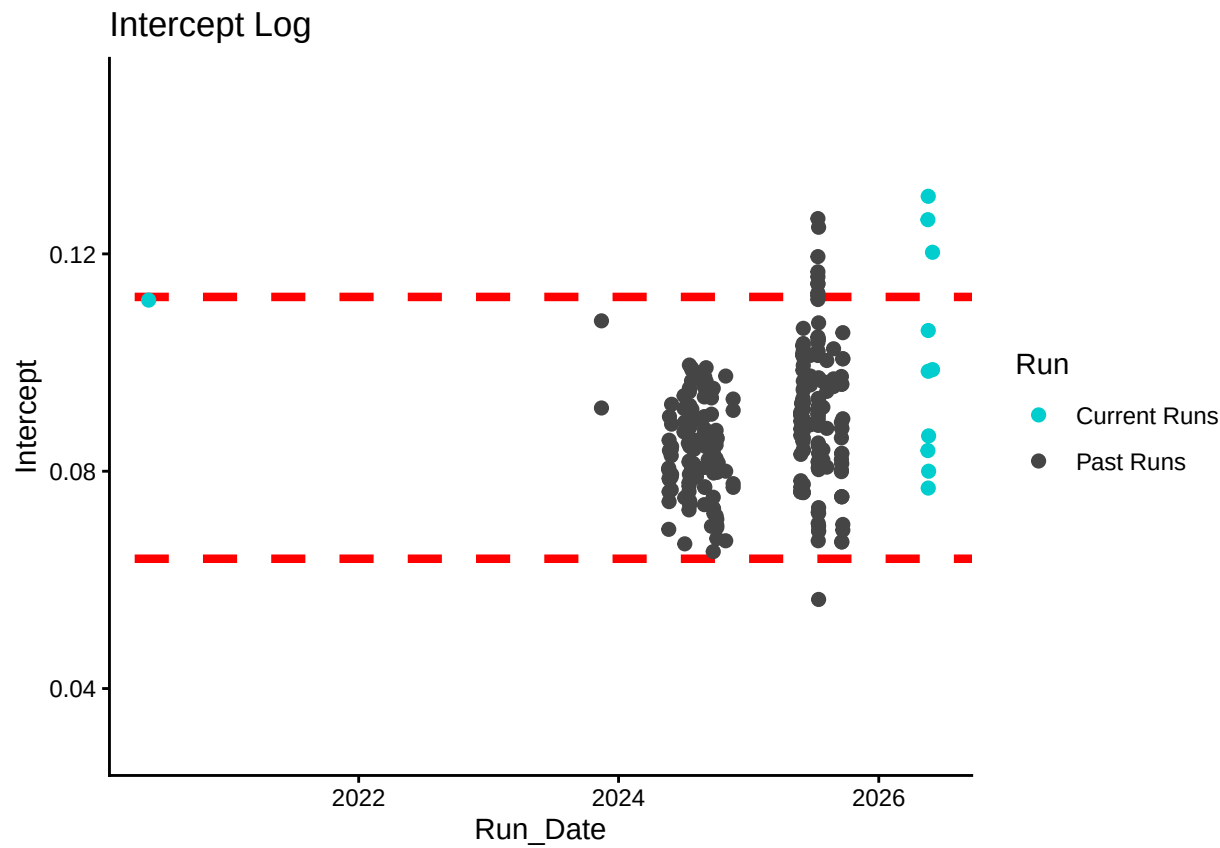
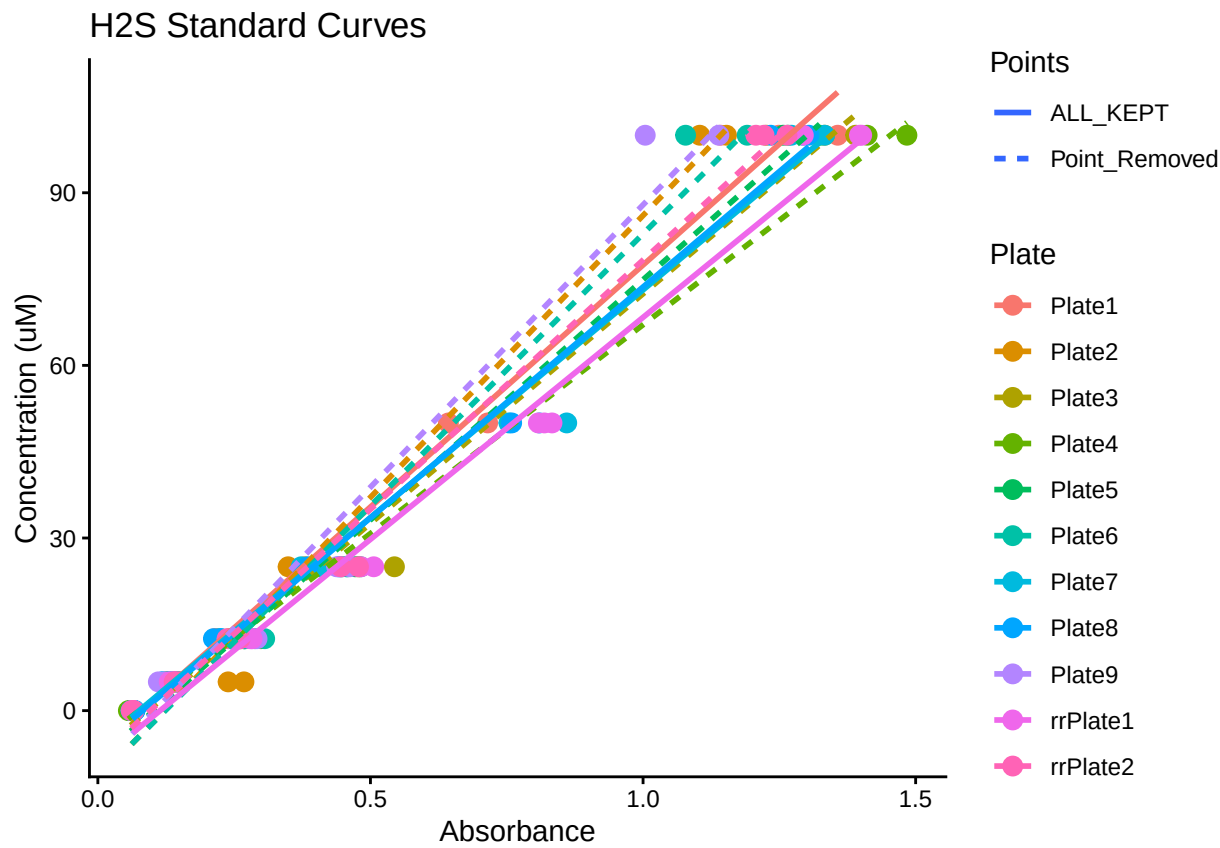
```
log_path <- "Processed Data/SMARTX_H2S_QAQC_2026.csv"

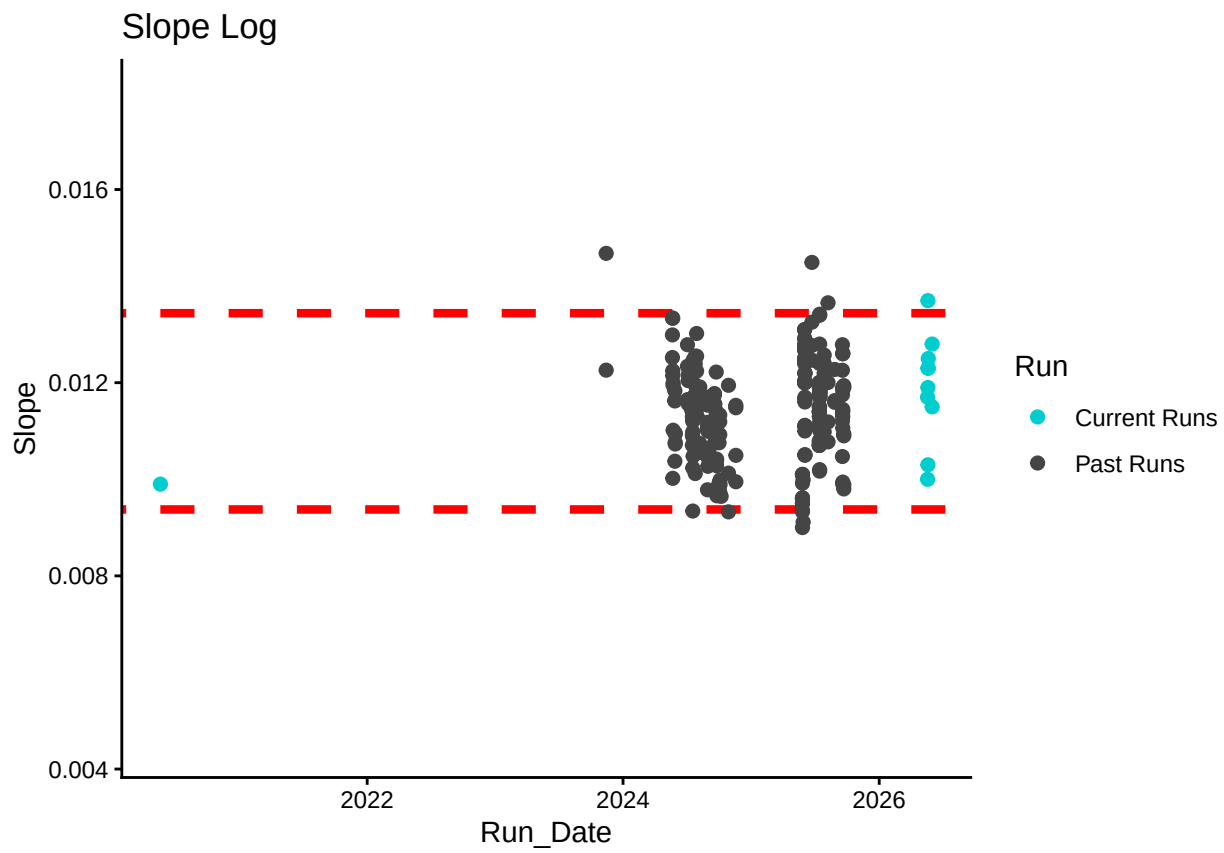
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202605.csv"
rerun_path <- "Processed Data/GCReW_SMARTX_Porewater_H2S_202605_reruns.csv"

##Read in metadata and create similar sample IDs for matching to samples
##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves





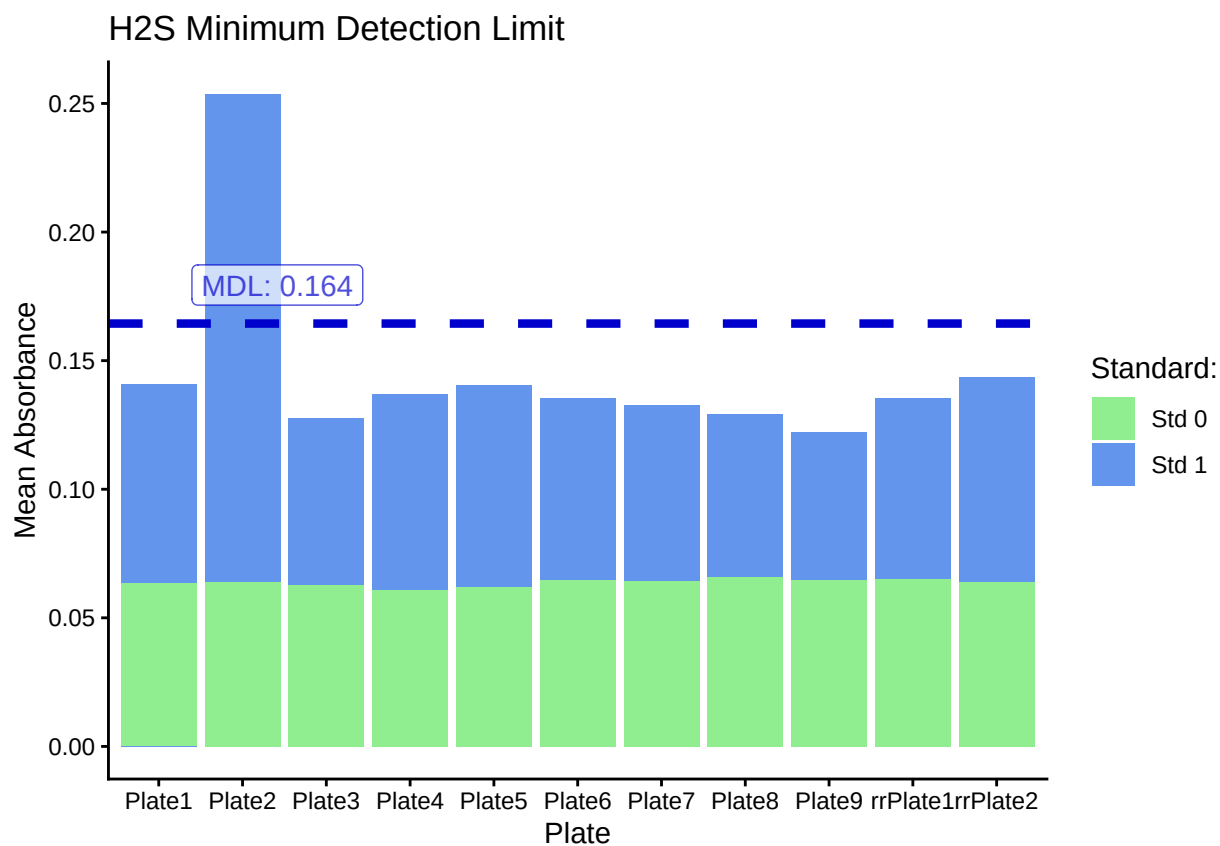
### Return Flags For Standard Curve

| Plate    | R2_Flag | CV_Flag |
|----------|---------|---------|
| Plate1   | R2_Good | CV_Good |
| Plate2   | R2_Bad  | CV_Good |
| Plate3   | R2_Bad  | CV_Good |
| Plate4   | R2_Good | CV_Good |
| Plate5   | R2_Good | CV_Good |
| Plate6   | R2_Bad  | CV_Good |
| Plate7   | R2_Good | CV_Good |
| Plate8   | R2_Good | CV_Good |
| Plate9   | R2_Bad  | CV_Good |
| rrPlate1 | R2_Good | CV_Good |
| rrPlate2 | R2_Good | CV_Good |

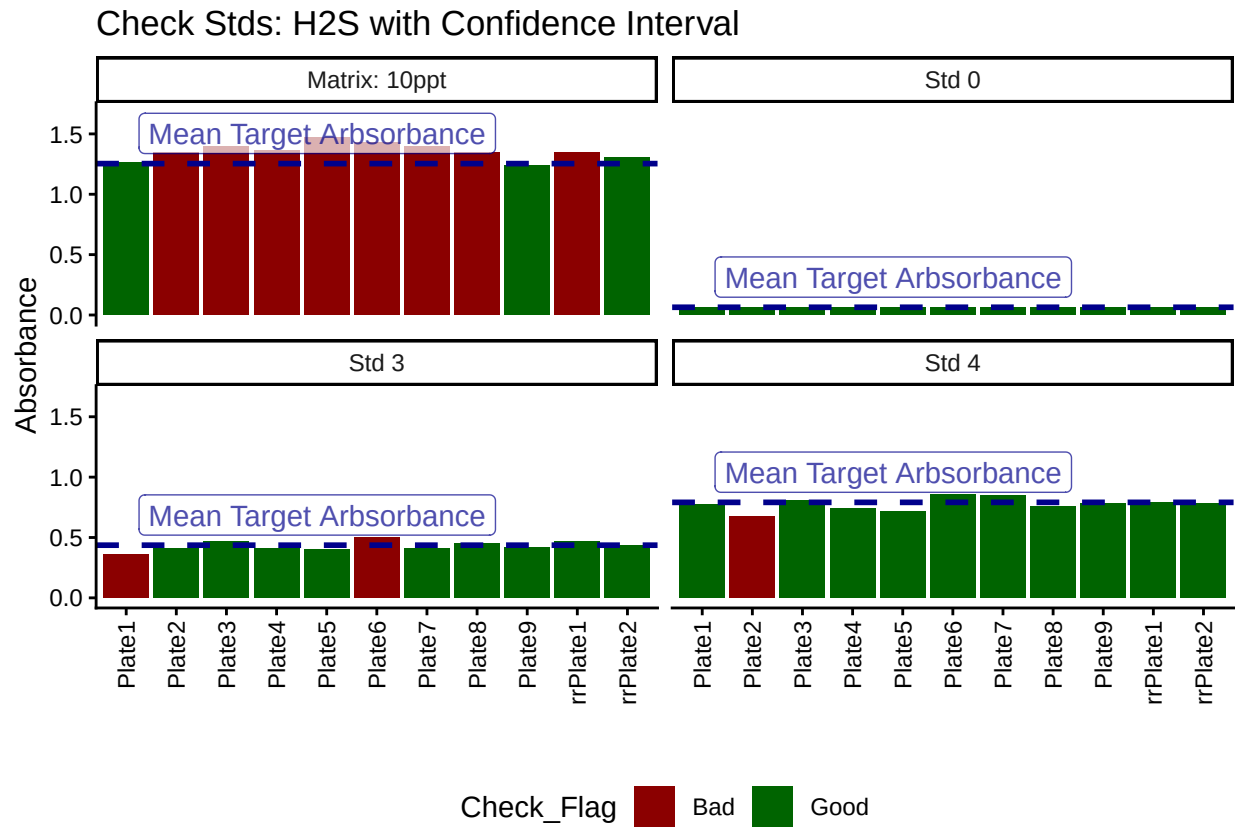
| Plate    | Curve_RMVD |
|----------|------------|
| Plate1   | KEPT       |
| Plate2   | REMOVED    |
| Plate3   | REMOVED    |
| Plate4   | REMOVED    |
| Plate5   | KEPT       |
| Plate6   | REMOVED    |
| Plate7   | KEPT       |
| Plate8   | KEPT       |
| Plate9   | REMOVED    |
| rrPlate1 | REMOVED    |
| rrPlate2 | KEPT       |

| Plate | Curve_RMVD |
|-------|------------|
|-------|------------|

## Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

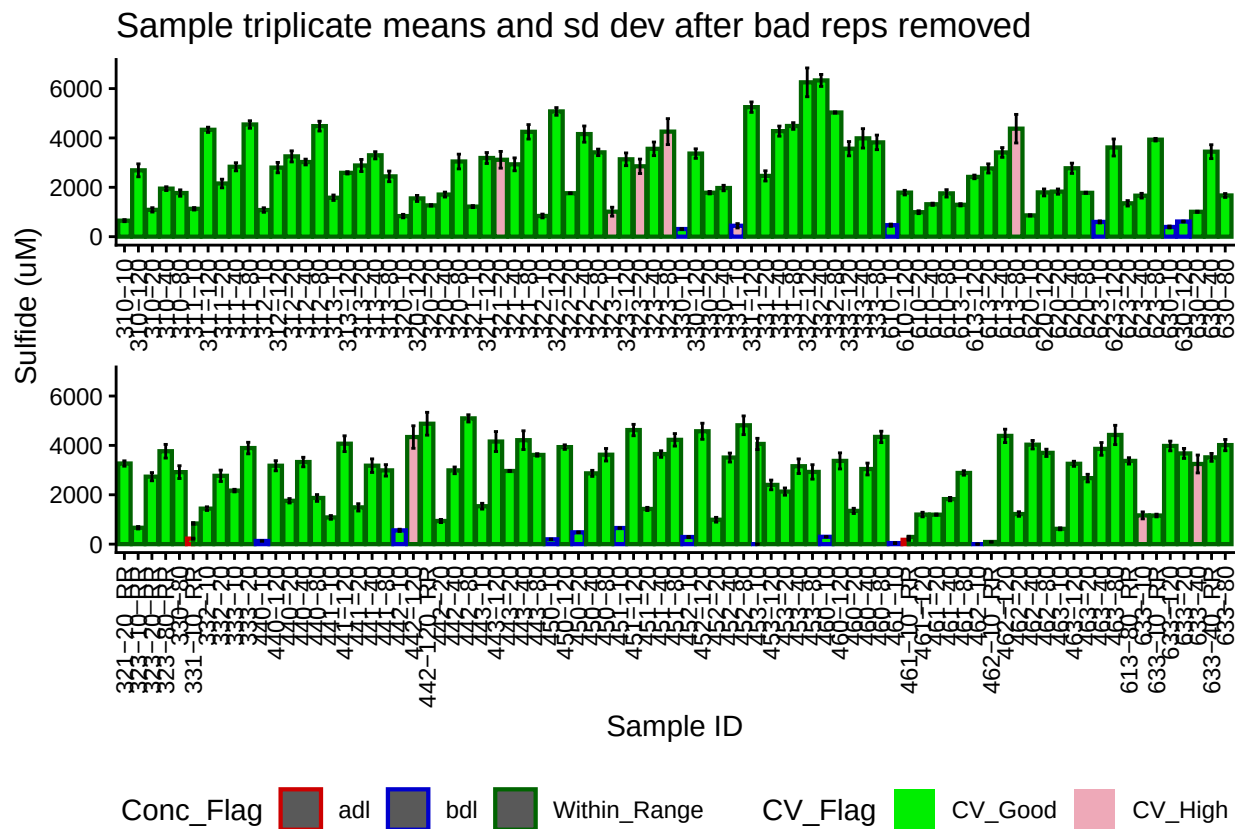
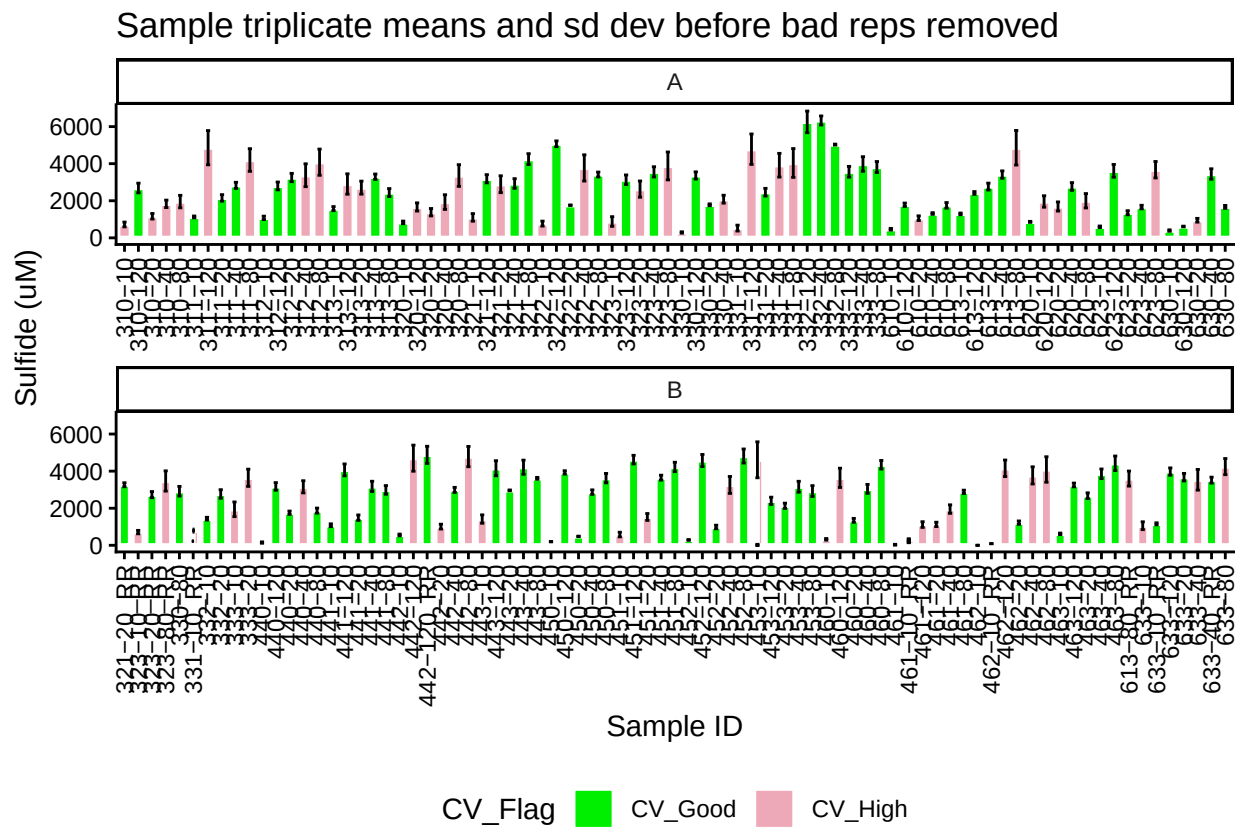
| Plate    | CHKs_Flag   |
|----------|-------------|
| Plate1   | Std3Chk_Bad |
| Plate2   | Std5Chk_Bad |
| Plate2   | Std4Chk_Bad |
| Plate3   | Std5Chk_Bad |
| Plate4   | Std5Chk_Bad |
| Plate5   | Std5Chk_Bad |
| Plate6   | Std3Chk_Bad |
| Plate6   | Std5Chk_Bad |
| Plate7   | Std5Chk_Bad |
| Plate8   | Std5Chk_Bad |
| rrPlate1 | Std5Chk_Bad |

| Plate | CHK_CV_Flags |
|-------|--------------|
|-------|--------------|

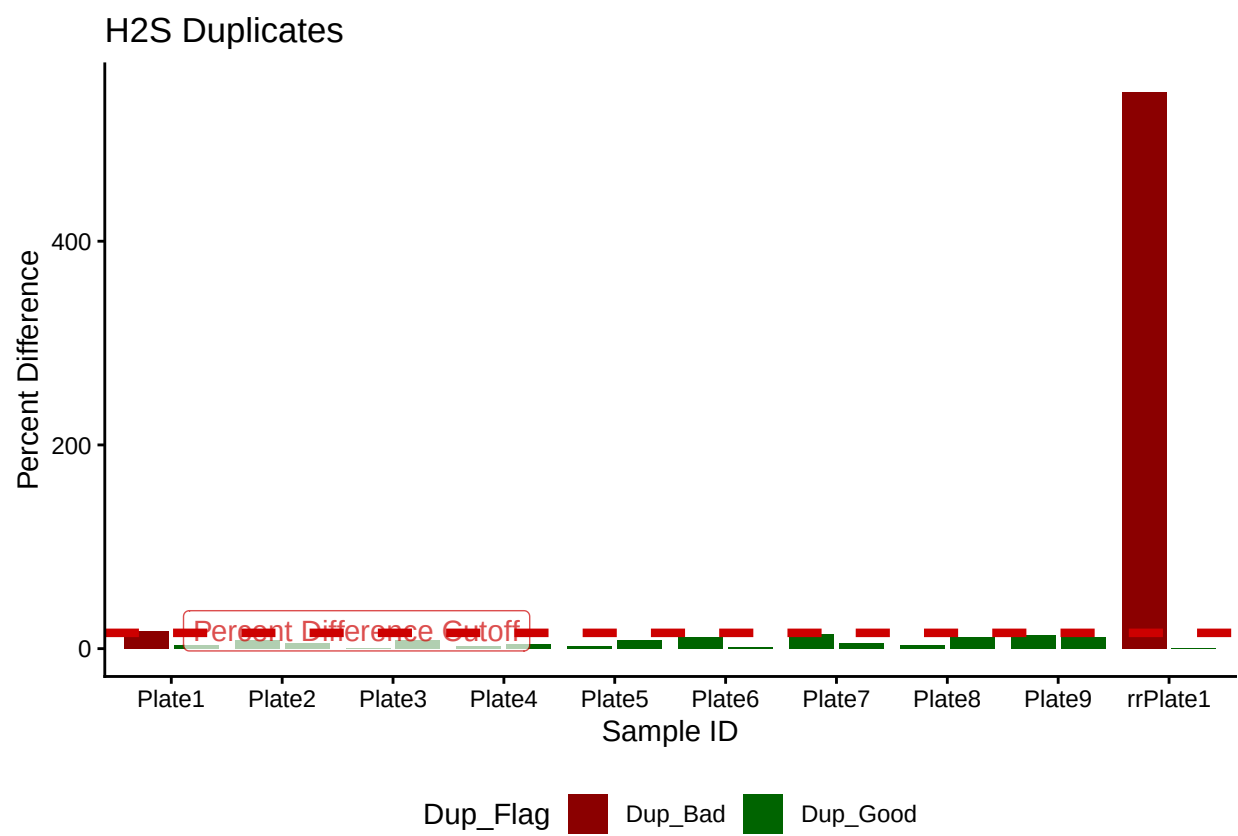
##Calculate Sulfide Concentrations & Add Concentration Flags

##Remove High CV Samples and Average

Plot Samples



Duplicate QAQC

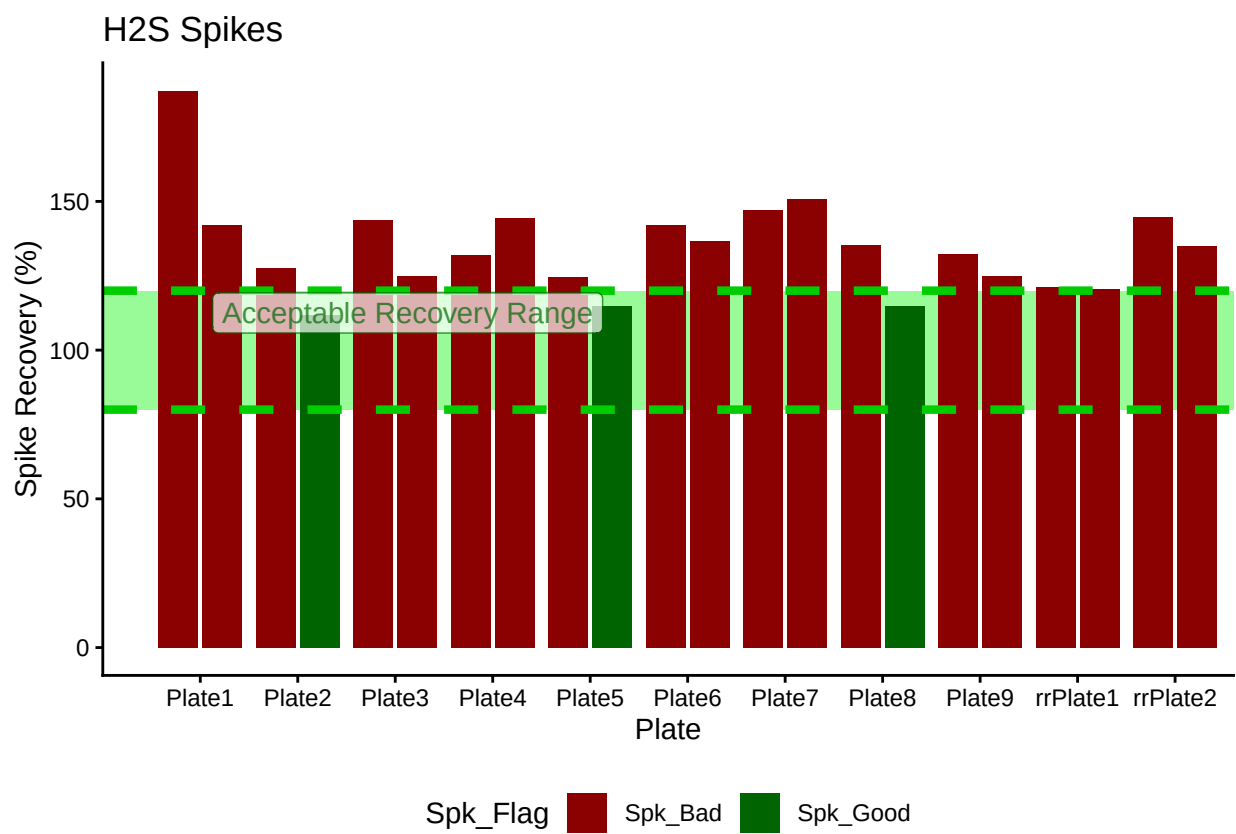


Display Dup Flags

## [1] ">60% Dups Pass"

| Plate    | Dup_Flag |
|----------|----------|
| Plate1   | Dup_Bad  |
| Plate2   | Dup_Good |
| Plate3   | Dup_Good |
| Plate4   | Dup_Good |
| Plate5   | Dup_Good |
| Plate6   | Dup_Good |
| Plate7   | Dup_Good |
| Plate8   | Dup_Good |
| Plate9   | Dup_Good |
| rrPlate1 | Dup_Bad  |

Spike QAQC



Display Spike Flags

```
## [1] "Spikes Paired"
```

| Plate    | Spike_Flag |
|----------|------------|
| Plate1   | Spk_Bad    |
| Plate2   | Spk_Bad    |
| Plate3   | Spk_Bad    |
| Plate4   | Spk_Bad    |
| Plate5   | Spk_Bad    |
| Plate6   | Spk_Bad    |
| Plate7   | Spk_Bad    |
| Plate8   | Spk_Bad    |
| Plate9   | Spk_Bad    |
| rrPlate1 | Spk_Bad    |
| rrPlate2 | Spk_Bad    |

```
## [1] "<60% of Spikes Pass, Rerun"
```

Merge samples with Metadata & check all samples are present

```
## All sample IDs are present in data
```

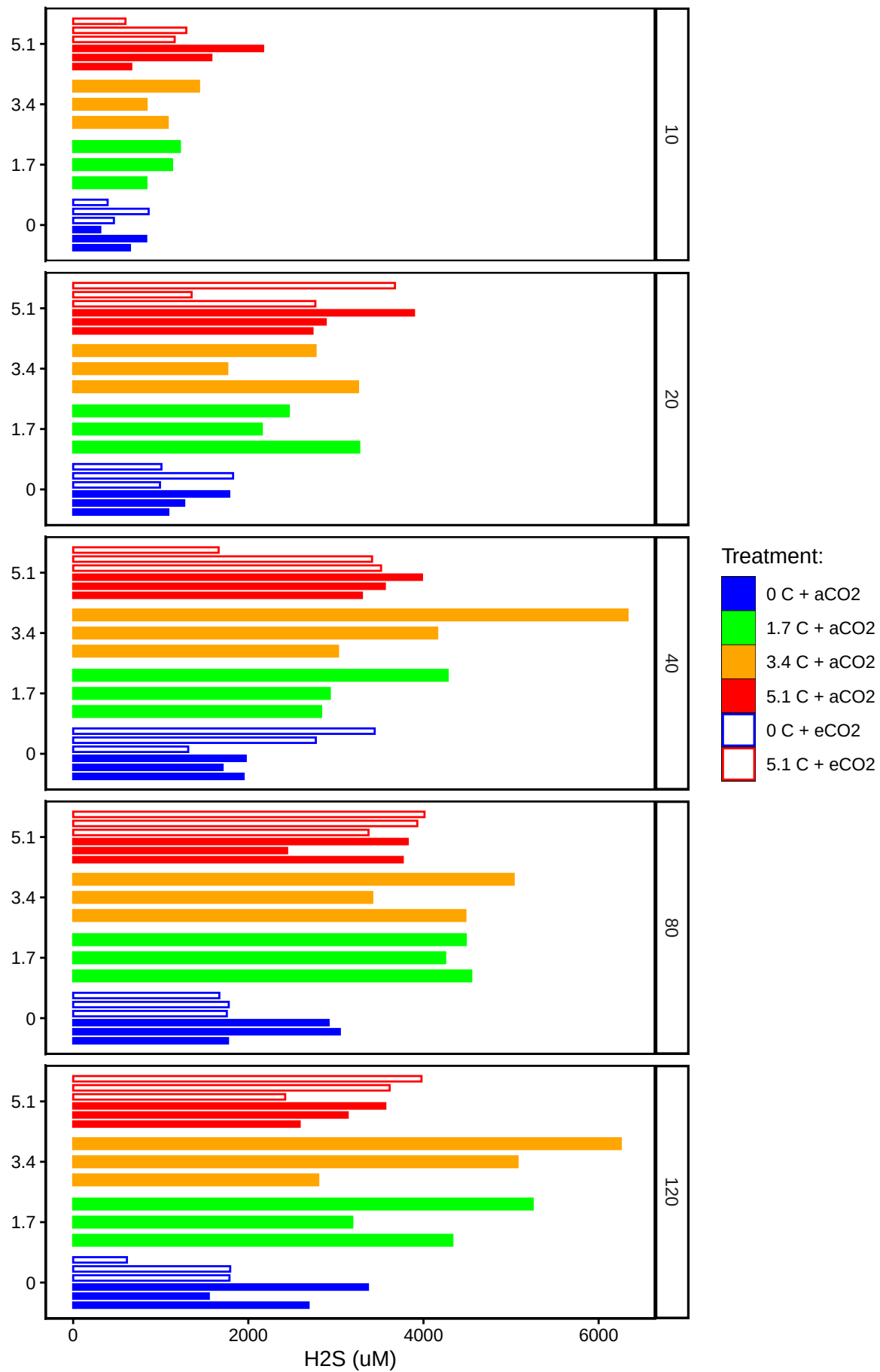
```
##Add Flags
```

```
##Organize Data
```

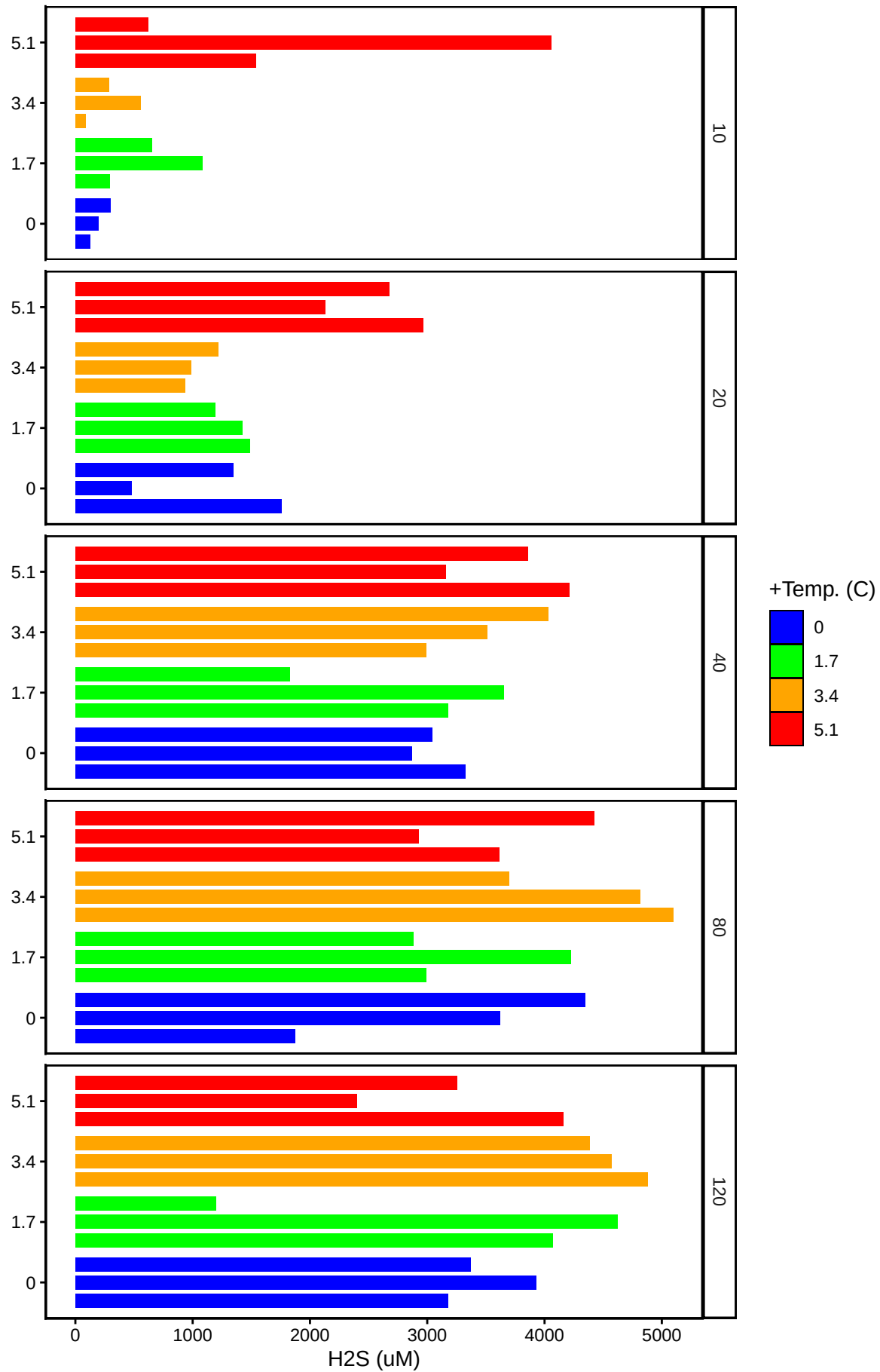
## Visualize Data

```
## Visualize Data
```

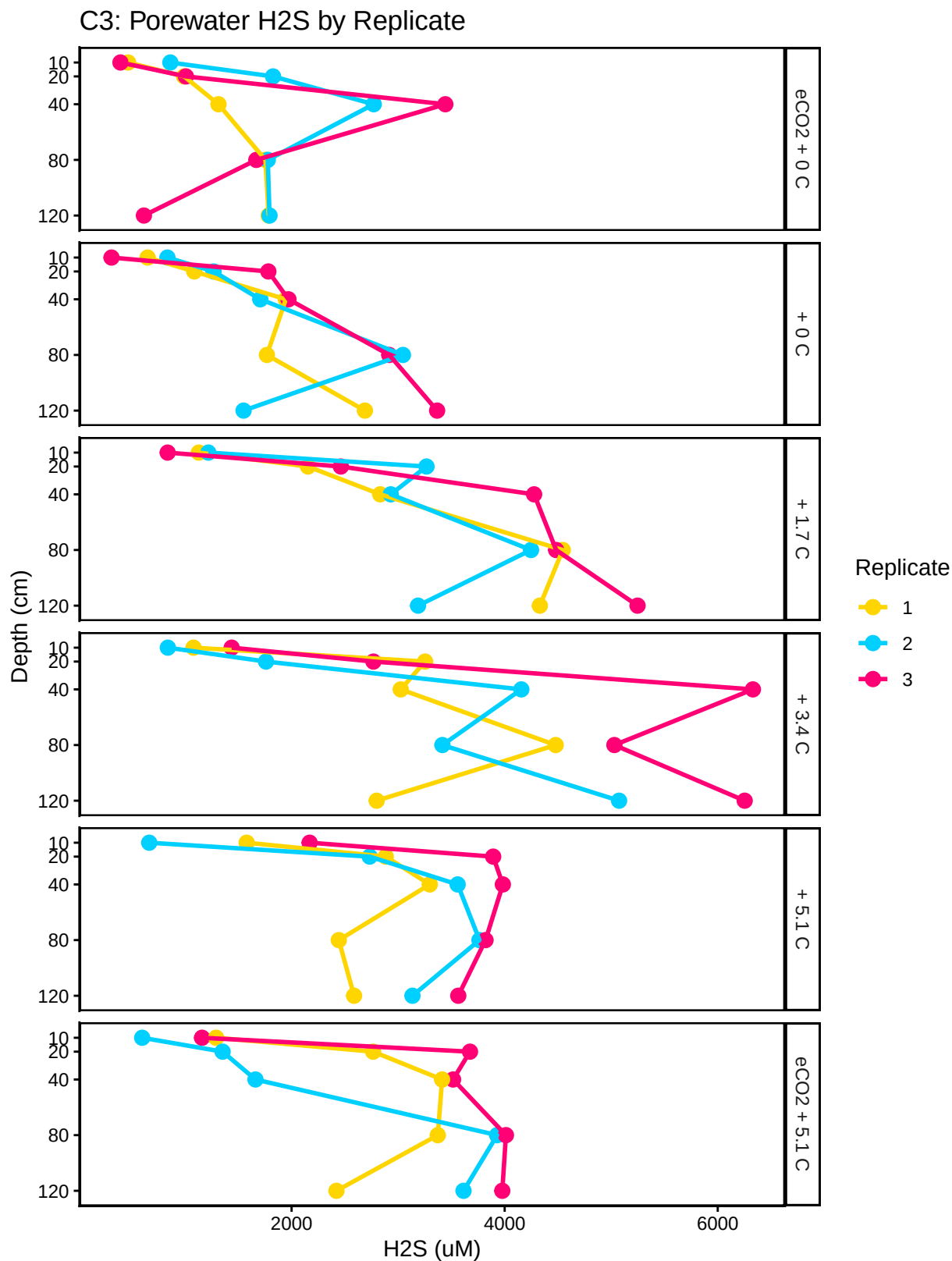
### C3: Porewater H2S



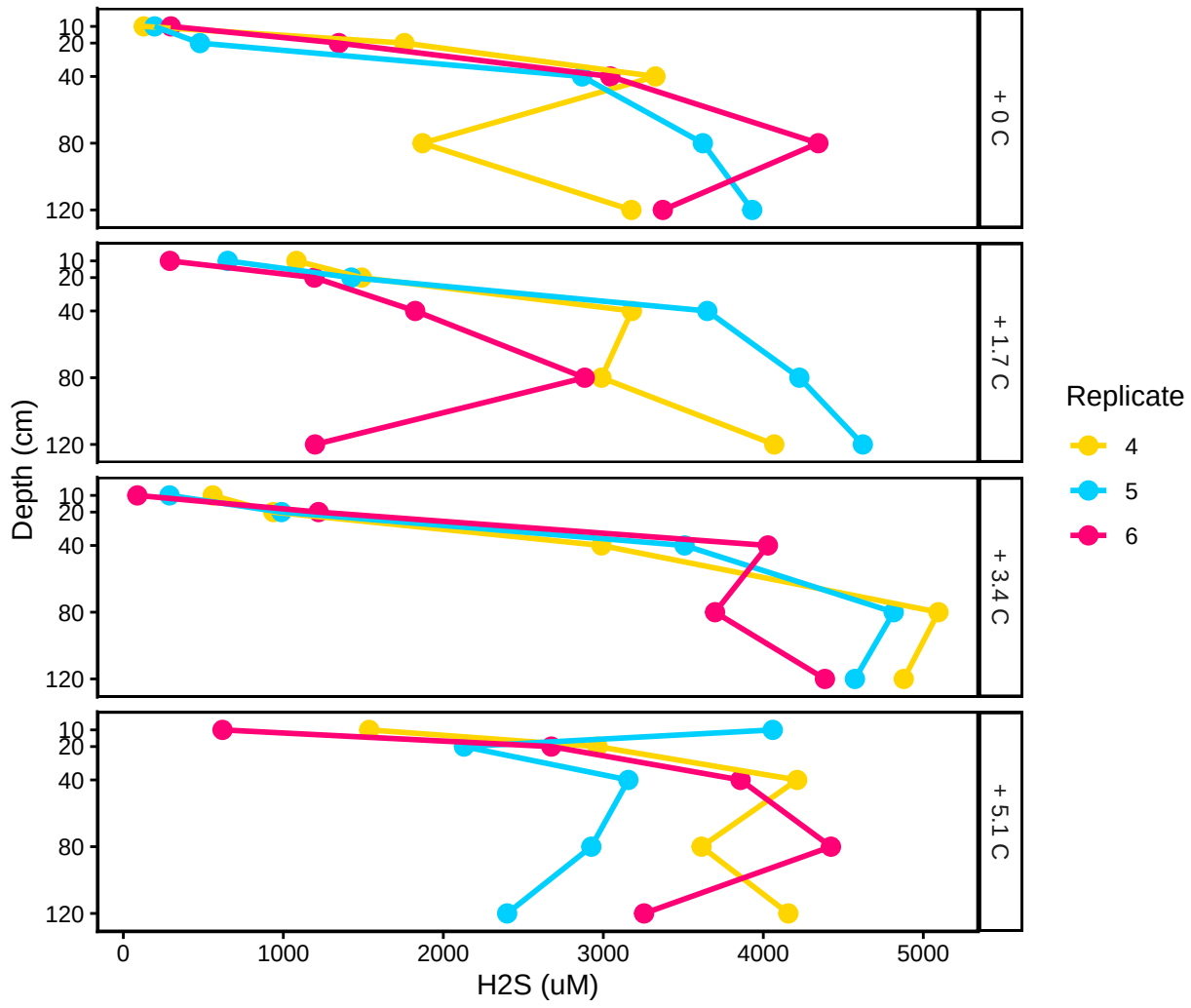
# C4: Porewater H2S



```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



C4: Porewater H<sub>2</sub>S by Replicate



##Export Data

END